The mode of operation is critical to the security of the encryption process.

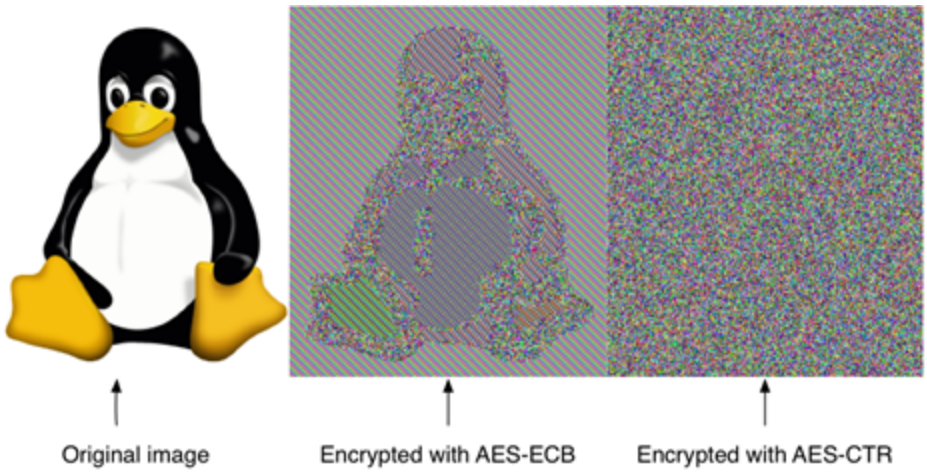


Figure 6.4 An image of the Linux mascot, Tux, that has been encrypted by AES in ECB mode. The shape of the penguin and many features are still visible despite the encryption. By contrast, the same image encrypted with AES in CTR mode is indistinguishable from random noise. Original image by Larry Ewing and The GIMP (https://commons.wikimedia.org/wiki/File:Tux.svg).

6.3.1 Authenticated encryption

Many encryption algorithms only ensure the confidentiality of data that has been encrypted and don’t claim to protect the integrity of that data. This means that an attacker won’t be able to read any sensitive attributes in an encrypted token, but they may be able to alter them. For example, if you know that a token is encrypted with CTR mode and (when decrypted) starts with the string `user=brian`, you can